

PREFERENCES—WARM-UP PROBLEM SET

ECON101—INTRODUCTION TO MICROECONOMICS.

SPRING 2026

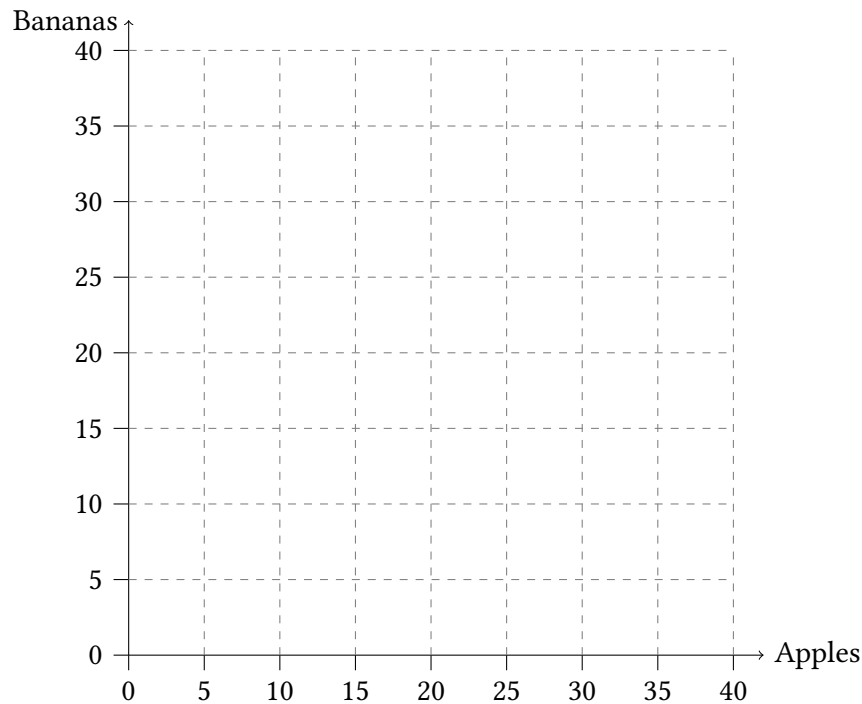
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Problem 1

Bagdat likes both apples and bananas. He consumes nothing else. The consumption bundle where Bagdat consumes x_A bushels of apples per year and x_B bushels of bananas per year is written as (x_A, x_B) . Last year, Bagdat consumed 20 bushels of apples and 5 bushels of bananas. It happens that the set of consumption bundles (x_A, x_B) such that Bagdat is indifferent between (x_A, x_B) and $(20, 5)$ is the set of all bundles such that $x_B = \frac{100}{x_A}$. The set of bundles (x_A, x_B) such that Bagdat is just indifferent between (x_A, x_B) and the bundle $(10, 15)$ is the set of bundles such that $x_B = \frac{150}{x_A}$.

- (a) On the graph below, plot several points that lie on the indifference curve that passes through the point $(20, 5)$, and sketch this curve, using blue ink. Do the same, using red ink, for the indifference curve passing through the point $(10, 15)$.
- (b) Use pencil to shade in the set of commodity bundles that Bagdat weakly prefers to the bundle $(10, 15)$. Use blue ink to shade in the set of commodity bundles such that Bagdat weakly prefers $(20, 5)$ to these bundles.



For each of the following statements about Bagdat's preferences, write "true" or "false."

- (c) $(30, 5) \sim (10, 15)$.

(d) $(10, 15) \succ (20, 5)$.

(e) $(20, 5) \succeq (10, 10)$.

(f) $(24, 4) \succeq (11, 9.1)$.

(g) $(11, 14) \succ (2, 49)$.

A set is convex if for any two points in the set, the line segment between them is also in the set.

(h) Is the set of bundles that Bagdat weakly prefers to $(20, 5)$ a convex set?

(i) Is the set of bundles that Bagdat considers inferior to $(20, 5)$ a convex set?

The slope of Bagdat's indifference curve through a point, (x_A, x_B) , is known as his marginal rate of substitution at that point.

(j) Remember that Bagdat's indifference curve through the point $(10, 10)$ has the equation $x_B = \frac{100}{x_A}$. Those of you who know calculus will remember that the slope of a curve is just its derivative, which in this case is $-\frac{100}{x_A^2}$. (If you don't know calculus, you will have to take our word for this.) Find Bagdat's marginal rate of substitution at the point, $(10, 10)$.

(k) What is his marginal rate of substitution at the point $(5, 20)$?

(l) What is his marginal rate of substitution at the point $(20, 5)$?

(m) Do the indifference curves you have drawn for Bagdat exhibit diminishing marginal rate of substitution?

Problem 2

Nazerke consumes only nuts and berries. Fortunately, she likes both goods. The consumption bundle where Nazerke consumes x_1 units of nuts per week and x_2 units of berries per week is written as (x_1, x_2) . The set of consumption bundles (x_1, x_2) such that Nazerke is indifferent between (x_1, x_2) and $(1, 16)$ is the set of bundles such that $x_1 \geq 0, x_2 \geq 0$, and $x_2 = 20 - 4\sqrt{x_1}$. The set of bundles (x_1, x_2) such that $(x_1, x_2) \sim (36, 0)$ is the set of bundles such that $x_1 \geq 0, x_2 \geq 0$, and $x_2 = 24 - 4\sqrt{x_1}$.

(a) On the graph below, plot several points that lie on the indifference curve that passes through the point $(1, 16)$, and sketch this curve, using blue ink. Do the same, using red ink, for the indifference curve passing through the point $(36, 0)$.

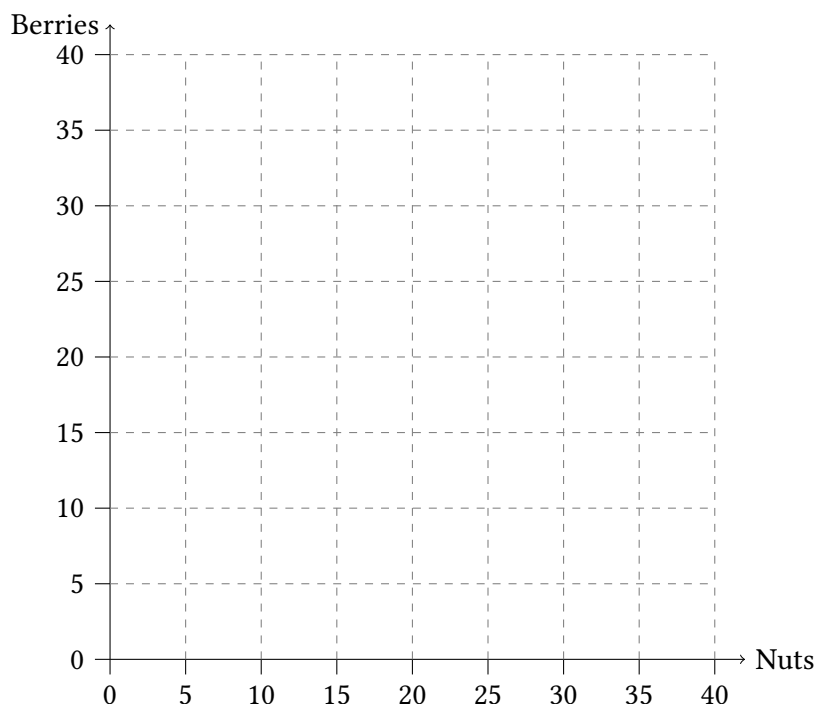
(b) Use pencil to shade in the set of commodity bundles that Nazerke weakly prefers to the bundle $(1, 16)$. Use red ink to shade in the set of all commodity bundles (x_1, x_2) such that Nazerke weakly prefers $(36, 0)$ to these bundles. Is the set of bundles that Nazerke prefers to $(1, 16)$ a convex set?

(c) What is the slope of Nazerke's indifference curve at the point $(9, 8)$? (Hint: Recall from calculus the way to calculate the slope of a curve. If you don't know calculus, you will have to draw your diagram carefully and estimate the slope.)

(d) What is the slope of her indifference curve at the point $(4, 12)$?

(e) What is the slope of her indifference curve at the point $(9, 12)$? What about at the point $(4, 16)$?

- (f) Do the indifference curves you have drawn for Nazerke exhibit diminishing marginal rate of substitution?
- (g) Does Nazerke have convex preferences?



Problem 3

Daniyar finds himself at a vending machine on a hot and dusty Sunday. The vending machine requires exact change—one 50 tenge coin and one 20 tenge coin. No other combination of coins will make anything come out of the machine. No stores are open; no one is in sight. Daniyar is so thirsty that the only thing he cares about is how many drinks he will be able to buy with the change in his pocket; the more he can buy, the better. While Daniyar searches his pockets, your task is to draw some indifference curves that describe Daniyar’s preferences about what he finds.

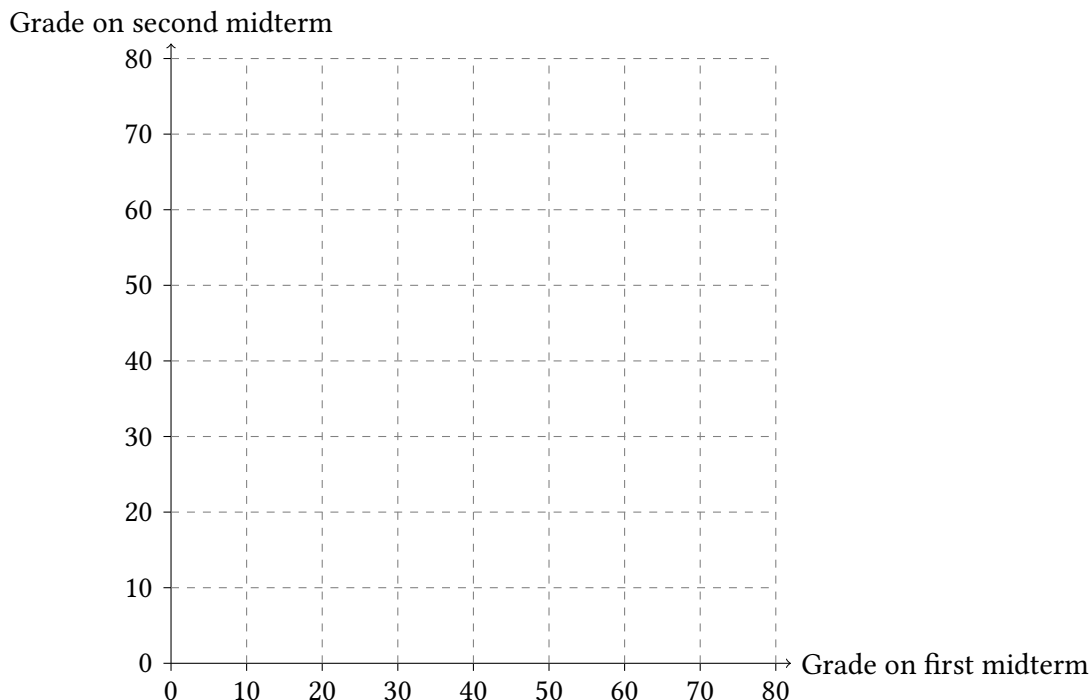
- (a) If Daniyar has 1 fifty-tenge coin and 1 twenty-tenge coin in his pockets, he can buy 1 drink. How many drinks can he buy if he has 2 fifty-tenge coins and 2 twenty-tenge coins?
- (b) Use red ink to shade in the area on the graph consisting of all combinations of fifty-tenge and twenty-tenge coins that Daniyar thinks are just indifferent to having 1 fifty-tenge coin and 1 twenty-tenge coin. (Imagine that it is possible for Daniyar to have fractions of fifty-tenge or twenty-tenge coins, but, of course, they would be useless in the machine.) Now use blue ink to shade in the area consisting of all combinations that Daniyar thinks are just indifferent to having 2 fifty-tenge coins and 2 twenty-tenge coins. Notice that Daniyar has indifference “bands,” not indifference curves.
- (c) Does Daniyar have convex preferences between twenty-tenge and fifty-tenge coins?
- (d) Does Daniyar always prefer more of both kinds of money to less?

- (e) Does Daniyar have a bliss point?
- (f) If Daniyar had arrived at the vending machine on a Saturday, the shop across the street would have been open. This shop has a soda fountain that will sell you as much soda as you want at a price of 10 tenge per 100 milliliters. The salesperson will take any combination of fifty-tenge and twenty-tenge coins in payment. Suppose that Daniyar plans to spend all of the money in his pocket on soda at the shop on Saturday. On the graph above, use pencil or black ink to draw one or two of Daniyar's indifference curves between fifty-tenge and twenty-tenge coins in his pocket. (For simplicity, draw your graph as if Daniyar's fractional fifty-tenge and fractional twenty-tenge coins are accepted at the corresponding fraction of their value.) Describe these new indifference curves in words.

Problem 4

Professor Legkovato always gives two midterms in his business class. He only uses the higher of the two scores that a student gets on the midterms when he calculates the course grade.

- (a) **Nazima Bilimdi** wants to maximize her grade in this course. Let x_1 be her score on the first midterm and x_2 be her score on the second midterm. Which combination of scores would Nazima prefer, $x_1 = 20$ and $x_2 = 70$ or $x_1 = 60$ and $x_2 = 60$?
- (b) On the graph below, use red ink to draw an indifference curve showing all of the combinations of scores that Nazima likes exactly as much as $x_1 = 20$ and $x_2 = 70$. Also use red ink to draw an indifference curve showing the combinations that Nazima likes exactly as much as $x_1 = 60$ and $x_2 = 60$.
- (c) Does Nazima have convex preferences over these combinations?



Nazima is also taking a course in economics from Professor AlekHARDov. Professor AlekHARDov gives two midterms. Instead of discarding the lower grade, Professor AlekHARDov discards the higher one.

- (d) Let x_1 be Nazima's score on the first midterm and x_2 be her score on the second midterm. Which combination of scores would Nazima prefer, $x_1 = 20$ and $x_2 = 70$ or $x_1 = 60$ and $x_2 = 50$?
- (e) On the graph above, use blue ink to draw an indifference curve showing all of the combinations of scores on her econ exams that Nazima likes exactly as well as $x_1 = 20$ and $x_2 = 70$. Also use blue ink to draw an indifference curve showing the combinations that Nazima likes exactly as well as $x_1 = 60$ and $x_2 = 50$.
- (f) Does Nazima have convex preferences over these combinations?

Problem 5

Coach Anabolinov likes his players to be big, fast, and obedient. If player A is better than player B in two of these three characteristics, then Coach Anabolinov prefers A to B , but if B is better than A in two of these three characteristics, then Anabolinov prefers B to A . Otherwise, Anabolinov is indifferent between them. Wilbur weighs 340 pounds, runs very slowly, and is fairly obedient. Harold weighs 240 pounds, runs very fast, and is very disobedient. Jerry weighs 150 pounds, runs at average speed, and is extremely obedient.

- (a) Does Anabolinov prefer Wilbur to Harold or vice versa?
- (b) Does Anabolinov prefer Harold to Jerry or vice versa?
- (c) Does Anabolinov prefer Wilbur to Jerry or vice versa?
- (d) Does Coach Anabolinov have transitive preferences?
- (e) After several losing seasons, Coach Anabolinov decides to change his way of judging players. According to his new preferences, Anabolinov prefers player A to player B if player A is better in all three of the characteristics that Anabolinov values, and he prefers B to A if player B is better at all three things. He is indifferent between A and B if they weigh the same, are equally fast, and are equally obedient. In all other cases, Coach Anabolinov simply says " A and B are not comparable."
- (f) Are Coach Anabolinov's new preferences complete?
- (g) Are Coach Anabolinov's new preferences transitive?